Some Extra Optimization Problems

1. What are the dimensions of the most economical cylindrical can having capacity of $100\,cm^3$?

2. Determine the area of the largest rectangle that can be inscribed in a right triangle with legs adjacent to the right angle having lengths of 10cm and 12cm. The two sides of the rectangle lie along the legs.

3. At 1:00pm a merchant ship is sailing 18 km/hr due east. At the same time a patrol boat 40 km south of the merchant ship is travelling north at 24 km/hr. When will they be closest to each other?

4. A powerhouse, P is on one bank of a straight river 200 m wide and a factory, F is on the other side of the bank 400 m downstream from P. A power cable is suppose to connect P with F. The cost to lay the power cable underwater is $12/m and the cost to lay the power cable underground is $6/m. How should the power cable be laid to minimize cost?